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AUTHENTICATION SERVER, AUTHENTICATION SYSTEM, CONTROL METHOD OF AUTHENTICATION SERVER, AND STORAGE MEDIUM

Abstract

An authentication server includes a user registration unit, a service registration unit, and a storage unit. The user registration unit acquires a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user. The service registration unit processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider. The service registration unit generates a third ID that is uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider. The storage unit stores the first biological information, the first ID, the second ID, and the third ID in association with each other.

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Background/Summary

[0001] This application is a Continuation of U.S. application Ser. No. 17/916, 113 filed on Sep. 30, 2022, which is a National Stage Entry of PCT/JP2020/016177 filed on Apr. 10, 2020, the contents of all of which are incorporated herein by reference, in their entirety.

TECHNICAL FIELD

[0002] The present invention relates to an authentication server, an authentication system, a control method of an authentication server, and a storage medium.

BACKGROUND ART

[0003] In recent years, various kinds of services using biological information have begun to spread. For example, face authentication is used for various kinds of procedures (check-in procedures, procedures at baggage drop-off counters, etc.) performed in airports and for check-in procedures at hotels, etc.

[0004] In the case of a service using face authentication, processing is performed as follows. First, a terminal (a terminal installed at an airport or a hotel) acquires a face image of a user and generates feature values (a feature vector) that characterize the face image. The generated feature values are transmitted to a server on a network.

[0005] The server includes a database that stores biological information and personal information (a name, an address, etc.) of the user who wishes to receive the service using face authentication. When the server acquires a matching request from the terminal, the server searches the above database (performs matching) and determines the biological information and the personal information corresponding to the matching request from the terminal. The server transmits the determined personal information to the terminal, and the terminal installed at the airport or the like performs an operation based on the acquired personal information.

[0006] For example, PTL 1 discloses a private lodging management server in a private lodging service system. This management server performs identification of a lodger by using an image of the guest captured by a mobile terminal and unlocks a room door.

CITATION LIST

Patent Literature

[0007] [PTL 1] Japanese Unexamined Patent Application Publication No. JP2018-101235

SUMMARY OF INVENTION

Technical Problem

[0008] FIG. 2 in PTL 1 illustrates a configuration in which the server that performs authentication based on biological information manages personal information in association with the biological information. This configuration could cause a serious problem if information is leaked from the database in the server. In particular, biological information of a person, such as a face image, a fingerprint image, and a vein pattern, does not change for his or her entire life. That is, biological information cannot be changed even if information leakage occurs. Leakage of a combination of such biological information and personal information to a third party could cause an irreversible situation.

[0009] As measures against the above information leakage, personal information or the like can be encrypted. However, if encrypted text and a key for decrypting the encrypted text are stored in the

same server, the key and the encrypted text could be leaked simultaneously. Thus, encrypting personal information or the like is not an ultimate solution.

[0010] It is a main object of the present invention to provide an authentication server, an authentication system, a control method of an authentication server, and a storage medium that enable more secure authentication using biological information.

Solution to Problem

[0011] According to a first aspect of the present invention, there is provided an authentication server including: a user registration unit that acquires a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; a service registration unit that processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; and a storage unit, wherein the service registration unit generates a third ID that is uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider, and wherein the storage unit stores the first biological information, the first ID, the second ID, and the third ID in association with each other.

[0012] According to a second aspect of the present invention, there is provided an authentication system including: an authentication server; and a management server, wherein the authentication server includes a user registration unit that acquires a first ID that uniquely determines a user in the system and first biological information that is used for authentication of the user, a service registration unit that processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider, and a first storage unit, wherein the service registration unit generates a third ID that is uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider, wherein the first storage unit stores the first biological information, the first ID, the second ID, and the third ID in association with each other, and wherein the management server includes a personal information acquisition unit that acquires the first ID and personal information of the user from the user, a service registration request unit that acquires the third ID by transmitting the service registration request to the authentication server, and a second storage unit that stores personal information of the user and the third ID in association with each other.

[0013] According to a third aspect of the present invention, there is provided a control method of an authentication server, the control method including: acquiring a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; receiving a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; generating a third ID that is uniquely determined by a combination of the user and the service provider; transmitting the third ID to the service provider; and storing the first biological information, the first ID, the second ID, and the third ID in association with each other.

[0014] According to a fourth aspect of the present invention, there is provided a computer-readable storage medium storing a program causing a computer mounted on an authentication server to perform processing for: acquiring a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; receiving a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; generating a third ID that is uniquely determined by a combination of the user and the service provider; transmitting the third ID to the service provider; and storing the first biological information, the first ID, the second ID, and the third ID in association with each other.

Advantageous Effects of Invention

[0015] The individual aspects of the present invention provide an authentication server, an authentication system, a control method of an authentication server, and a storage medium that

enable more secure authentication using biological information. The advantageous effects of the present invention are not limited to the above advantageous effect. The present invention may provide other advantageous effects, instead of or in addition to the above advantageous effect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a diagram illustrating an outline of an example embodiment.

[0017] FIG. 2 is a diagram illustrating an example of a schematic configuration of an authentication system according to a first example embodiment.

[0018] FIG. 3 is a diagram illustrating an operation in a user registration phase in the authentication system according to the first example embodiment.

[0019] FIG. 4 is a diagram illustrating an operation in a service registration phase in the authentication system according to the first example embodiment.

[0020] FIG. 5 is a diagram illustrating an operation in a service provision phase in the authentication system according to the first example embodiment.

[0021] FIG. 6 is a diagram illustrating an example of a processing configuration of an authentication server according to the first example embodiment.

[0022] FIG. 7 is a diagram illustrating an operation of a user registration unit in the authentication server according to the first example embodiment.

[0023] FIG. 8 is a diagram illustrating an operation of the user registration unit in the authentication server according to the first example embodiment.

[0024] FIG. 9 is a diagram illustrating an example of an authentication information database.

[0025] FIG. 10 is a diagram illustrating an example of the authentication information database.

[0026] FIG. 11 is a diagram illustrating an example of the authentication information database.

[0027] FIG. 12 is a diagram illustrating an example of a processing configuration of a management server according to the first example embodiment.

[0028] FIG. 13 is a diagram illustrating an operation of a personal information acquisition unit in the management server according to the first example embodiment.

[0029] FIG. 14 is a diagram illustrating an example of a user information database.

[0030] FIG. 15 is a diagram illustrating an example of a processing configuration of an authentication terminal according to the first example embodiment.

[0031] FIG. 16 is a sequence diagram illustrating an example of an operation relating to the service registration phase in the authentication system according to the first example embodiment.

[0032] FIG. 17 is a sequence diagram illustrating an example of an operation relating to the service provision phase in the authentication system according to the first example embodiment.

[0033] FIG. 18 is a diagram illustrating an operation in a service registration phase in an authentication system according to a second example embodiment.

[0034] FIG. 19 is a diagram illustrating an operation in a service provision phase in the authentication system according to the second example embodiment.

[0035] FIG. 20 is a diagram illustrating an example of an authentication information database.

[0036] FIG. 21 is a diagram illustrating an operation in a service registration phase in an authentication system according to a third example embodiment.

[0037] FIG. 22 is a diagram illustrating an operation in a service provision phase in the authentication system according to the third example embodiment.

[0038] FIG. 23 is a diagram illustrating an example of a user information database.

[0039] FIG. 24 is a diagram illustrating an example of an authentication information database.

[0040] FIG. 25 is a diagram illustrating an example of a hardware configuration of an authentication server.

EXAMPLE EMBODIMENT

[0041] First, an outline of an example embodiment will be described. In the following outline, various components are denoted by reference characters for the sake of convenience. That is, the following reference characters are used as examples to facilitate the understanding of the present invention. Thus, the description of the outline is not intended to impose any limitations. In addition, unless otherwise specified, an individual block illustrated in the drawings represents a configuration of a functional unit, not a hardware unit. An individual connection line between blocks in the drawings signifies both one-way and two-way directions. An arrow schematically illustrates a principal signal (data) flow and does not exclude bidirectionality. In the present description and drawings, elements that can be described in a like way will be denoted by a like reference character, and redundant description thereof will be omitted as needed.

[0042] An authentication server **100** according to an example embodiment includes a user registration unit **101**, a service registration unit **102**, and a storage unit **103** (see FIG. 1). The user registration unit **101** acquires a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user. The service registration unit **102** processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider. The service registration unit **102** generates a third ID uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider. The storage unit **103** stores the first biological information, the first ID, the second ID, and the third ID in association with each other.

[0043] While biological information of users and various kinds of IDs are stored in the storage unit **103** in the authentication server **100**, personal information (for example, names, etc.) of the users are not stored in the storage unit **103**. Thus, even if information is leaked from the authentication server **100**, because no personal information associated with biological information is obtained by a third party, the participants (the system users and service providers) in the authentication system can use the authentication based on the authentication server **100** more securely. By storing the personal information of an individual user and a corresponding third ID in association with each other, the service provider can acquire personal information necessary for service provision from a third ID acquired from the authentication server **100**.

[0044] Hereinafter, specific example embodiments will be described in more detail with reference to drawings.

First Example Embodiment

[0045] A first example embodiment will be described in more detail with reference to drawings.

[System Configuration]

[0046] FIG. 2 is a diagram illustrating an example of a schematic configuration of an authentication system according to a first example embodiment. As illustrated in FIG. 2, the authentication system includes an authentication center and a plurality of service providers.

[0047] Each service provider that participates in the authentication system provides a service using biological authentication. Examples of the services provided by the service providers include payment services at retail stores, etc. and accommodation services at hotels, etc. Examples of the services provided by the service providers also include emigration and immigration examination at airports and ports. The service providers disclosed in the present application may provide any services that can be provided by using biological authentication.

[0048] An authentication server **10** is installed in the authentication center. The authentication server **10** operates as an authentication center station that performs authentication using biological information. The authentication server **10** may be a server installed on the premises of the authentication center or may be a server installed on a cloud.

[0049] Examples of the biological information of a user include data (feature values) calculated from physical features unique to this individual, such as a face, a fingerprint, a voiceprint, a vein, a

retina, or an iris pattern of an eye of the user. Alternatively, the biological information of a user may be image data of a face image or a fingerprint image of the user, for example. The biological information of a user may be any information including physical features of the user.

[0050] The authentication server **10** is a server apparatus for realizing an individual service based on biological authentication. The authentication server **10** processes an “authentication request” transmitted from an individual service provider and transmits a corresponding result of the authentication processing to the service provider.

[0051] The individual service provider has a management server and an authentication terminal.

[0052] For example, a management server **20** and a plurality of authentication terminals **30** are installed at a service provider S1. A management server **20** and a plurality of authentication terminals **31** are installed at a service provider S2.

[0053] In the following description, when a certain component needs to be distinguished from the other components, a reference numeral on the right side of the corresponding hyphen will be used. As the apparatuses included in the service provider S1 operate in the same way as those included in the service provider S2, the following description will be made with a focus on the service provider S1.

[0054] The apparatuses illustrated in FIG. 2 are connected to each other. For example, the authentication server **10** and the individual management server **20** are each connected to wired or wireless communication means so that the authentication server **10** and the individual management server **20** can communicate with each other.

[0055] The individual management server **20** is a server that controls and manages overall operations of the corresponding service provider. For example, if a service provider is a retail store, the corresponding management server **20** performs merchandise inventory control, etc. If the service provider is a hotel operator, the corresponding management server **20** manages information about guest reservations, for example.

[0056] In addition to the above functions relating to the provision of a service, the individual management server **20** has control and management functions relating to biological authentication of the users.

[0057] The individual authentication terminal **30** is an apparatus that functions as an interface for the users who visit at the corresponding service provider. The users receive various services via the authentication terminal **30**. For example, if the service provider is a retail store, a user performs payment by using the authentication terminal **30**. If the service provider is a hotel operator, a user performs a check-in procedure by using the authentication terminal **30**.

[0058] The configuration illustrated in FIG. 2 is an example and does not limit the configuration or the like of the authentication system disclosed in the present application. For example, two or more authentication servers **10** may be installed in the authentication center. At least one authentication terminal **30** is installed by the individual service provider. Alternatively, the functions of the management server **20** and the functions of the individual authentication terminal **30** may be integrated, and in this case, a single apparatus having the integrated functions may be installed to provide a service using biological authentication. At each service provider, a plurality of authentication terminals **30** may be connected to a single management server **20** as illustrated in FIG. 2 or a single authentication terminal **30** may alternatively be connected to a single management server **20**.

[Schematic System Operation]

[0059] Next, a schematic operation in the authentication system according to the first example embodiment will be described.

[0060] The operation in the authentication system includes three phases.

[0061] The first phase is a phase (a user registration phase) in which a user performs system registration.

[0062] The second phase is a phase (a service registration phase) in which the user performs

service registration.

[0063] The third phase is a phase (a service provision phase) in which a service using biological authentication is provided to the user.

[User Registration Phase]

[0064] FIG. 3 is a diagram illustrating an operation in the user registration phase in the authentication system according to the first example embodiment.

[0065] A user who wishes to receive a service using biological authentication performs user registration in advance. The user determines information (a user ID (Identifier) and a PW (password)) for determining himself or herself in the authentication system and registers the information in the system. In the drawings including FIG. 3, the user ID will be denoted as “uID”. [0066] In addition, the user registers his or her biological information (for example, a face image) in the system. The user registers the above three items of information (the user ID, the password, and the biological information) in the system by using any means. For example, the user may mail a document in which the above three items of information are written to the authentication center, and an employee at the authentication center may enter the above three items of information to the authentication server 10. Alternatively, the user may mail an external storage device such as a USB (Universal Serial Bus) memory in which the above three items of information are stored to the authentication center.

[0067] Alternatively, the user may enter his or her face image captured by operating his or her terminal 40, a user ID, and a password to the authentication server 10. Examples of the terminal 40 include a portable terminal device such as a smartphone, a portable phone, a game console, or a tablet and a computer (a personal computer or a laptop computer).

[0068] The authentication server 10 generates feature values (a feature vector formed by a plurality of feature values) from the acquired face image and stores the feature values, the user ID, and the password in association with each other. Specifically, the authentication server 10 adds a new entry in its authentication information database and stores the above three items of information in association with each other.

[0069] In this way, in the user registration phase, a first ID (for example, a user ID) that uniquely determines a user in the system and first biological information used for authentication of the user are registered in the system. In the first example embodiment, a user ID and a password are used as the identifier (the first ID) that uniquely determines a user in the system. However, if each user uses a unique user ID, the user ID may be used as the above identifier (the first ID).

[Service Registration Phase]

[0070] FIG. 4 is a diagram illustrating an operation in the service registration phase in the authentication system according to the first example embodiment.

[0071] Upon completing the user registration, the user selects a service provider from which the user wishes to receive a service using biological authentication and registers the selected service provider in the system. For example, in FIG. 2, if the user wishes to receive a service from the service provider S1, the user registers the service provider S1 in the system.

[0072] The user registers personal information (for example, his or her name, etc.) necessary for receiving a service from the selected service provider in the system. Examples of the personal information include his or her name, age, and gender. In addition to the above personal information, the user registers the user ID and the password determined in the user registration phase in the system.

[0073] In the disclosure of the present application, the personal information is defined as information that does not include biological information of the user (the authentication target user). That is, biological information and feature values generated from the biological information are excluded from “personal information” in the disclosure of the present application.

[0074] The user enters the above three items of information (the personal information, the user ID, and the password) to the selected service provider by using any means. For example, the user mails

a medium (a paper medium, an electronic medium) in which the above three items of information are written to the selected service provider. An employee of the service provider enters the above three items of information to the corresponding management server **20**. The user may enter the above three items of information to the corresponding management server **20** by operating an authentication terminal **30** installed at the service provider.

[0075] Alternatively, as illustrated in FIG. **4**, the user enters the above three items of information to the management server **20** by operating his or her terminal **40**. In this case, the user enters the above three items of information on a web page managed and operated by the service provider.

[0076] After the management server **20** acquires the above three items of information (the personal information, the user ID, and the password), the management server **20** transmits a “service registration request” to the authentication server **10**. Specifically, the management server **20** transmits a service registration request including a service provider ID, the user ID, and the password to the authentication server **10**.

[0077] The service provider ID is identification information for uniquely identifying the service provider (for example, a retail store participating in the authentication infrastructure using biological authentication) included in the authentication system. In the example in FIG. **2**, a different service provider ID is assigned to each of the service providers S1 and S2.

[0078] An individual service provider ID is an ID assigned to each service provider, not to each service. For example, in FIG. **2**, even if the service providers S1 and S2 provide the same kind of service (for example, an accommodation service), if the service providers S1 and S2 are different business operators, different IDs are assigned to these service providers.

[0079] The authentication server **10** and the management server **20** share the service provider ID by using any method. For example, when a service provider participates in the authentication infrastructure, the authentication server **10** can generate a service provider ID and distribute (transmit) the generated service provider ID to the service provider. In the drawings including FIG. **4**, the service provider ID will be denoted as “spID”.

[0080] Upon receiving the service registration request, the authentication server **10** searches its authentication information database by using the user ID and the password included in the request as a key and determines the corresponding user. Thereafter, the authentication server **10** generates a “service user ID”.

[0081] The service user ID is identification information that uniquely identifies the correspondence relationship between (a combination of) the user and the service provider. For example, in the example in FIG. **2**, a value that is set as a service user ID defined by a combination of a user U1 and the service provider S1 and a value that is set as a service user ID defined by a combination of the user U1 and the service provider S2 are different values.

[0082] The authentication server **10** stores the user ID, the password, the feature values, the service provider ID, and the above-generated service user ID in association with each other. In the drawings including FIG. **4**, the service user ID will be denoted as “suID”.

[0083] The authentication server **10** transmits the above-generated service user ID to the management server **20** that has transmitted the service registration request. The authentication server **10** gives the service user ID to the management server **20** by transmitting a response including the service user ID to the management server **20**.

[0084] The management server **20** stores the service user ID acquired from the authentication server **10** and the personal information of the user in association with each other. The management server **20** adds a new entry in a user information database and stores the above information (the personal information and the service user ID). The user repeats the registration operation as described above for each service provider from which the user wishes to receive a service using biological authentication. In other words, the user does not perform the registration operation for service providers from which the user does not wish to receive services.

[0085] As described above, in the service registration phase, the service provider of a service that

the user wishes to use transmits a service registration request including a first ID (for example, the user ID) and a second ID (for example, the service provider ID) to the authentication server **10**. When processing the service registration request, the authentication server **10** generates a third ID (for example, the service user ID) uniquely determined by a combination of the user and the service provider. The authentication server **10** transmits the third ID to the service provider. The service provider (the management server **20**) stores the personal information of the user and the third ID in association with each other.

[Service Provision Phase]

[0086] FIG. **5** is a diagram illustrating an operation in the service provision phase in the authentication system according to the first example embodiment.

[0087] Upon completion of the registration of a service (the service registration phase), the user visits the corresponding service provider. The user stands in front of an authentication terminal **30**.

[0088] The authentication terminal **30** acquires biological information from the user standing in front of the authentication terminal **30**. Specifically, the authentication terminal **30** captures an image of the user and acquires a face image. The authentication terminal **30** transmits the acquired face image to the management server **20**.

[0089] The management server **20** generates feature values from the acquired face image. The management server **20** transmits an authentication request including the generated feature values and its service provider ID to the authentication server **10**.

[0090] The authentication server **10** extracts the feature values from the authentication request and performs matching processing (1-to-N processing; N will hereinafter denote a positive integer) by using the extracted feature values and the feature values registered in the authentication information database.

[0091] The authentication server **10** determines the user by performing the matching processing and determines the service user ID corresponding to the service provider ID included in the authentication request among the plurality of service user IDs associated with the determined user.

[0092] The authentication server **10** transmits the determined service user ID to the management server **20** that has transmitted the authentication request. The authentication server **10** transmits a response including the determined service user ID (a response to the authentication request) to the management server **20**.

[0093] The management server **20** searches its user information database by using the acquired service user ID as a key and determines the personal information corresponding to the service user ID. The service provider (the management server **20**, the authentication terminal **30**) provides the user with a service based on the determined personal information (for example, a payment procedure, a check-in procedure, etc.).

[0094] As described above, in the service provision phase, the authentication server **10** receives an authentication request including second biological information of a user and a second ID (a service provider ID) from a service provider. The authentication server **10** determines a third ID (a service user ID) by using the first and second biological information and the second ID. The authentication server **10** transmits the determined third ID to the service provider. When providing a service to the user, the management server **20** determines the personal information of the user by using the third ID acquired by transmitting the authentication request to the authentication server **10**.

[0095] The service provider provides a service to the user by using the determined personal information.

[0096] Next, details of the individual apparatuses included in the authentication system according to the first example embodiment will be described.

[Authentication Server]

[0097] FIG. **6** is a diagram illustrating an example of a processing configuration (processing modules) of the authentication server **10** according to the first example embodiment. As illustrated in FIG. **6**, the authentication server **10** includes a communication control unit **201**, a user

registration unit **202**, a database management unit **203**, a service registration unit **204**, an authentication unit **205**, and a storage unit **206**.

[0098] The communication control unit **201** is means for controlling communication with other apparatuses. Specifically, the communication control unit **201** receives data (packets) from the management servers **20**. In addition, the communication control unit **201** transmits data to the management servers **20**. The communication control unit **201** gives data received from other apparatuses to other processing modules. The communication control unit **201** transmits data acquired from other processing modules to other apparatuses. In this way, other processing modules transmit and receive data to and from other apparatuses via the communication control unit **201**.

[0099] The user registration unit **202** is means for realizing the above user registration. The user registration unit **202** acquires the user ID, the password, and the biological information (a face image) of a user (a user who wishes to receive a service using biological authentication; a system user).

[0100] The user registration unit **202** acquires the above three items of information (the user ID, the password, and the biological information) by using any means. For example, the user registration unit **202** displays a GUI (Graphical User Interface) or an input form on which the user ID and the password are determined on a terminal **40**. For example, the user registration unit **202** displays a GUI as illustrated in FIG. 7 on the terminal **40**.

[0101] The user registration unit **202** determines whether the user ID and password acquired on the GUI or the like have already been registered. If the user ID and password have not been registered yet, the user registration unit **202** displays a GUI for acquiring biological information of the user on the terminal **40**.

[0102] For example, the user registration unit **202** displays a GUI as illustrated in FIG. 8 on the terminal **40**. For example, the user presses a “Select File” button illustrated in FIG. 8 and specifies the image data of a face image to be registered in the system. The specified face image is displayed on a preview area (displayed as “Selected Face Image” in FIG. 8). To register a previewed face image, the user presses the “OK” button.

[0103] For example, the user registration unit **202** acquires the user ID, the password, and the biological information (a face image) on the GUIs as illustrated in FIGS. 7 and FIG. 8 and generates feature values (a feature vector formed by a plurality of feature values) from the face image.

[0104] Specifically, the user registration unit **202** extracts feature points from the acquired face image. An existing technique can be used to extract the feature points, and therefore, detailed description thereof will be omitted. For example, the user registration unit **202** extracts the eyes, nose, mouth, etc. from the face image as the feature points. Next, the user registration unit **202** calculates the location of an individual feature point and the distance between feature points as feature values and generates a feature vector formed by the plurality of feature values (vector information that characterizes the face image).

[0105] The user registration unit **202** gives the user ID, the password, and the feature values generated as described above to the database management unit **203**.

[0106] The database management unit **203** is means for managing the authentication information database. The authentication information database stores information (a user ID and a password) that determines a system user, biological information (feature values) of the user, a service provider ID that determines a service provider, and a service user ID that determines the user in a corresponding service in association with each other.

[0107] When the database management unit **203** acquires the above three items of information (the user ID, the password, and the feature values) from the user registration unit **202**, the database management unit **203** adds a new entry in the authentication information database. For example, when the database management unit **203** acquires the above three items of information about the

user U1, the database management unit **203** adds an entry illustrated in the lowest row in FIG. **9**. At the time of the user registration, since the service provider ID and the service user ID have not been generated yet, no information is set in these fields.

[0108] The service registration unit **204** is means for allowing a system user to register an individual service. The service registration unit **204** processes a service registration request acquired from the management server **20** of a service provider.

[0109] The service registration unit **204** searches the authentication information database by using the user ID and the password included in an acquired service registration request as a key. The service registration unit **204** checks the service provider ID field corresponding to the determined user (the user determined from a combination of the user ID and the password).

[0110] The service registration unit **204** determines whether the service provider ID included in the service registration request acquired from the management server **20** is set in the service provider ID field. If the service provider ID acquired from the management server **20** is already registered in the database, the service registration unit **204** notifies the management server **20** to that effect. In this case, since the service (the service provider) that the user wishes to register is already registered in the authentication information database, the service registration unit **204** transmits a “negative response” as a response to the service registration request.

[0111] In contrast, if the service provider ID included in the service registration request is not set in the service provider ID field corresponding to the determined user, the service registration unit **204** generates a service user ID corresponding to the user and the service provider.

[0112] As described above, a service user ID is identification information uniquely determined from a combination of a user and a service provider. For example, the service registration unit **204** calculates a hash value by using a user ID, a password, and a service provider ID and uses the calculated hash value as a service user ID. Specifically, the service registration unit **204** calculates a combined value of a user ID, a password, and a service provider ID, calculates a hash value of the combined value calculated, and uses the hash value as a service user ID.

[0113] The above generation of a service user ID by using a hash value is only an example. The method for generating a service user ID is not limited to the above generation method. A service user ID may be any information that can uniquely identify a combination of a system user and a service provider. For example, each time the service registration unit **204** processes a service registration request, the service registration unit **204** may generate a unique number and use this unique number as a service user ID.

[0114] After generating a service user ID, the service registration unit **204** gives the user ID, the password, the service provider ID, and the service user ID to the database management unit **203**. The database management unit **203** registers the two IDs (the service provider ID and the service user ID) in the authentication information database. For example, if the user U1 registers a service of the service provider S1, the above two IDs are added in an entry illustrated in the lowest row in FIG. **10**.

[0115] Since the service registration is performed per service provider, there are cases in which a plurality of service providers and a plurality of service user IDs are set for a single user. For example, if the user U1 registers a service of the service provider S1 and a service of the service provider S2, the entries in the second row and the third row in FIG. **11** are generated. If the user U2 registers a service of the service provider S1, an entry in the lowest row in FIG. **11** is generated.

[0116] The authentication information database illustrated in FIG. **11**, etc. is an example. The information stored in the authentication information database is not limited to the information illustrated in FIG. **11**, etc. For example, in place of feature values for authentication, a face image may be registered in the authentication information database. That is, at each authentication, feature values may be generated from a face image registered in the authentication information database.

[0117] When a service provider ID and a service user ID are registered in the authentication information database, the service registration unit **204** notifies the management server **20** that the

service registration request has been processed successfully. The service registration unit **204** transmits a “positive response” as a response to the service registration request. Specifically, the service registration unit **204** transmits a response including the service user ID to the management server **20**.

[0118] The authentication unit **205** is means for performing authentication processing for system users. The authentication unit **205** processes the authentication requests received from the management servers **20** of the service providers.

[0119] The authentication unit **205** extracts the feature values and the service provider ID from an authentication request. The authentication unit **205** searches the authentication information database by using the extracted feature values and service provider ID as a key and determines a corresponding service user ID.

[0120] The authentication unit **205** performs 1-to-N matching by setting the feature values extracted from an authentication request as the matching target feature values and by setting the feature values stored in the database as the registered feature values. Specifically, the authentication unit **205** calculates the similarity between a set of matching target feature values and each of a plurality of sets of registered feature values. For the individual similarity, the chi-squared distance, the Euclidean distance, or the like may be used. A longer distance represents a lower similarity, and a shorter distance represents a higher similarity.

[0121] The authentication unit **205** determines whether, among the plurality of sets of feature values registered in the database, there is at least one set of feature values whose similarity to the set of matching target feature values is a predetermined value or more. Next, if the authentication unit **205** finds a plurality of sets of feature values whose similarity to the set of matching target feature values is a predetermined value or more, the authentication unit **205** determines a set of feature values whose similarity is the highest among the plurality of sets of feature values. If the authentication unit **205** finds such set of feature values, the authentication unit **205** determines whether there is an entry matching the service provider ID included in the authentication request among the at least one service provider ID associated with the user, the at least one service provider ID having been determined by the above 1-to-N matching.

[0122] If there is an entry as described above (if the authentication unit **205** succeeds in the above two determination operations), the authentication unit **205** determines that the authentication of the user has succeeded. Specifically, the authentication unit **205** transmits a “positive response” to the management server **20** that has transmitted the authentication request. In this case, the authentication unit **205** generates a response including the service user ID in the determined entry (a response to the authentication request) and transmits the response to the management server **20**.

[0123] If at least one of the above two determination operations fails, the authentication unit **205** determines that the authentication of the user has failed. In this case, the authentication unit **205** transmits a “negative response” to the management server **20** that has transmitted the authentication request.

[0124] For example, in the example in FIG. **11**, if feature values “FV1” and a service provider ID “S1” are included in an authentication request, the entries (a user) in the second and third rows are first determined by the feature values FV1, and the entry in the second row is next determined by the service provider ID “S1”. As a result, the above authentication request is processed successfully, and a positive response including a service user ID “U1S1” is transmitted to the management server **20**.

[0125] In contrast, if feature values “FV2” and a service provider ID “S2” are included in an authentication request, the entry in the lowest row is determined by the feature values. However, since the service provider ID in the entry indicates “S1”, not “S2”, the above authentication request is not processed successfully. As a result, a negative response is transmitted to the management server **20**.

[0126] The storage unit **206** stores information necessary for operations of the authentication server

10. The authentication information database is established in the storage unit **206**.

[Management Server]

[0127] FIG. **12** is a diagram illustrating an example of a processing configuration (processing modules) of a management server **20** according to the first example embodiment. As illustrated in FIG. **12**, the management server **20** includes a communication control unit **301**, a personal information acquisition unit **302**, a service registration request unit **303**, a database management unit **304**, an authentication request unit **305**, and a storage unit **306**.

[0128] The communication control unit **301** is means for controlling communication with other apparatuses. Specifically, the communication control unit **301** receives data (packets) from the authentication server **10** and the corresponding authentication terminals **30**. In addition, the communication control unit **301** transmits data to the authentication server **10** and the corresponding authentication terminals **30**. The communication control unit **301** gives data received from other apparatuses to other processing modules. The communication control unit **301** transmits data acquired from other processing modules to other apparatuses. In this way, other processing modules transmit and receive data to and from other apparatuses via the communication control unit **301**.

[0129] The personal information acquisition unit **302** is means for acquiring personal information needed when the corresponding service provider provides a service. For example, when the service provider is a “retail store”, the personal information acquisition unit **302** acquires, in addition to the name of a user, etc., information about payment (for example, credit card information or bank account information). When the service provider is a “hotel operator”, the personal information acquisition unit **302** acquires, in addition to the name, etc., reservation information about accommodation (for example, accommodation date, etc.).

[0130] The personal information acquisition unit **302** acquires, in addition to the above personal information such as the name, the user ID and the password that the user has determined at the time of the system registration.

[0131] The personal information acquisition unit **302** acquires the personal information, the user ID, and the password by using any means. For example, the personal information acquisition unit **302** displays a GUI or a form for entering the above information on a terminal **40** (see FIG. **13**). Alternatively, information as illustrated in FIG. **13** may be displayed on a web page managed and operated by the corresponding service provider. Alternatively, a terminal **40** may download an application provided by the service provider, and the application may display a GUI or a form as illustrated in FIG. **13**. In particular, the web page may be a web page for managing information about the members of the service provider. That is, the members of an individual service provider may perform service registration on a web page for managing information about themselves.

[0132] The personal information acquisition unit **302** gives the personal information, the user ID, and the password acquired by using a GUI or the like to the service registration request unit **303**.

[0133] The service registration request unit **303** is means for requesting (asking) the authentication server **10** to register a service to be used by a user.

[0134] The service registration request unit **303** selects the user ID and the password from the above three items of information (the personal information, the user ID, and the password) acquired from the personal information acquisition unit **302**. The service registration request unit **303** transmits a service registration request including the selected user ID and password and the corresponding service provider ID to the authentication server **10**.

[0135] The service registration request unit **303** acquires a response to the service registration request from the authentication server **10**. If the acquired response is a “negative response”, the service registration request unit **303** notifies the user to that effect. For example, the service registration request unit **303** notifies the user that the service registration has already been performed.

[0136] If the acquired response is a “positive response”, the service registration request unit **303**

notifies the user that the service registration has succeeded. In addition, the service registration request unit **303** gives the service user ID included in the response and the personal information acquired from the personal information acquisition unit **302** to the database management unit **304**. [0137] The database management unit **304** is means for managing the user information database. The user information database is a database for managing information about the service provision target users (the system users). The user information database stores personal information (for example, the name, etc.) of an individual user and an individual service user ID acquired from the authentication server **10** in association with each other.

[0138] Upon acquiring the above information (the personal information and the service user ID) from the service registration request unit **303**, the database management unit **304** adds a new entry in the user information database. For example, when the management server **20** of the service provider S1 acquires the above information about the user U1, an entry illustrated in the lowest row in FIG. **14** is added.

[0139] The authentication request unit **305** is means for requesting the authentication server **10** to perform the authentication of the user.

[0140] Upon acquiring biological information (a face image) from a corresponding authentication terminal **30**, the authentication request unit **305** generates feature values from the face image. The authentication request unit **305** transmits an authentication request including the generated feature values and the corresponding service provider ID to the authentication server **10**.

[0141] If the response from the authentication server **10** is a “negative response” (if the authentication fails), the authentication request unit **305** notifies the authentication terminal **30** to that effect.

[0142] If the response from the authentication server **10** is a “positive response” (if the authentication succeeds), the authentication request unit **305** extracts the service user ID included in the response transmitted from the authentication server **10**. The authentication request unit **305** searches the user information database by using the service user ID as a key and determines a corresponding entry.

[0143] The authentication request unit **305** reads out the personal information set in the personal information field of the determined entry and transmits the personal information to the authentication terminal **30**. For example, in the example in FIG. **14**, if the service user ID is “U1S1”, the personal information in the lowest row is transmitted to the authentication terminal **30**.

[0144] The storage unit **306** stores information necessary for operations of the management server **20**. The user information database is established in the storage unit **306**.

[Authentication Terminal]

[0145] An authentication terminal **30** acquires personal information of a user from the corresponding management server **20** by transmitting biological information acquired from the user to the management server **20**. The authentication terminal **30** provides a service to the user by using the acquired personal information.

[0146] FIG. **15** is a diagram illustrating an example of a processing configuration (processing modules) of an authentication terminal **30** according to the first example embodiment. As illustrated in FIG. **15**, the authentication terminal **30** includes a communication control unit **401**, a biological information acquisition unit **402**, a service provision unit **403**, a message output unit **404**, and a storage unit **405**.

[0147] The communication control unit **401** is means for controlling communication with other apparatuses. Specifically, the communication control unit **401** receives data (packets) from the corresponding management server **20**. In addition, the communication control unit **401** transmits data to the corresponding management server **20**. The communication control unit **401** gives data received from other apparatuses to other processing modules. The communication control unit **401** transmits data acquired from other processing modules to other apparatuses. In this way, other processing modules transmit and receive data to and from other apparatuses via the communication

control unit **401**.

[0148] The biological information acquisition unit **402** is means for acquiring biological information (a face image) of a user by controlling a camera. The biological information acquisition unit **402** captures an image of a person in front of the authentication terminal **30** regularly or at predetermined timing. The biological information acquisition unit **402** determines whether a face image of the person is included in the acquired image. If a face image is included in the acquired image, the face image is extracted from the acquired image data.

[0149] An existing technique can be used for the face image detection and extraction processing performed by the biological information acquisition unit **402**, and therefore, detailed description thereof will be omitted. For example, the biological information acquisition unit **402** may extract a face image (a face area) from the image data by using a learning model learned by a CNN (Convolutional Neural Network). Alternatively, the biological information acquisition unit **402** may extract a face image by using a technique such as template matching.

[0150] The biological information acquisition unit **402** gives the extracted face image to the service provision unit **403**.

[0151] The service provision unit **403** is means for providing a predetermined service to the user. The service provision unit **403** transmits a face image acquired from the biological information acquisition unit **402** to the management server **20**. The management server **20** transmits, as a reply, personal information (for example, the name, etc.) corresponding to the face image. The service provision unit **403** provides a service to the user by using the personal information transmitted as a reply.

[0152] The message output unit **404** is means for outputting various messages to the user. For example, the message output unit **404** outputs a message relating to a user authentication result and a message relating to a service to be provided. The message output unit **404** may display a message by using a display device such as a liquid crystal monitor or plays an audio message by using an audio device such as a speaker.

[0153] The storage unit **405** stores information necessary for operations of the authentication terminal **30**.

[System Operation]

[0154] Next, operations in the authentication system according to the first example embodiment will be described. Hereinafter, while operations in the service registration phase and the service provision phase will be described, an operation in the user registration phase will be omitted.

[0155] FIG. **16** is a sequence diagram illustrating an example of an operation relating to the service registration phase in the authentication system according to the first example embodiment.

[0156] A management server **20** acquires personal information (information necessary for providing a service), a user ID, and a password from a user (step **S01**).

[0157] The management server **20** transmits a service registration request including the acquired user ID and password and its service provider ID to the authentication server **10** (step **S02**).

[0158] The authentication server **10** generates a service user ID by using the acquired user ID, password, and service provider ID (step **S03**).

[0159] The authentication server **10** stores the service provider ID and the service user ID in its authentication information database (step **S04**).

[0160] The authentication server **10** transmits a response including the service user ID (a response to the service registration request) to the management server **20** (step **S05**).

[0161] The management server **20** stores the personal information acquired in step **S01** and the service user ID acquired from the authentication server **10** in association with each other in its user information database (step **S06**).

[0162] FIG. **17** is a sequence diagram illustrating an example of an operation relating to the service provision phase in the authentication system according to the first example embodiment. Since the service registration phase has already been completed at the time of the service provision phase, the

authentication server **10** already holds, as the first biological information, the first feature values generated from a face image of the user.

[0163] An authentication terminal **30** acquires a face image (biological information) of the user and transmits the acquired face image to the management server **20** (step **S11**).

[0164] The management server **20** generates feature values (second feature values) from the acquired face image (step **S12**). The management server **20** handles the generated feature values as the second biological information.

[0165] The management server **20** transmits an authentication request including the generated feature values and its service provider ID to the authentication server **10** (step **S13**).

[0166] The authentication server **10** performs authentication processing by using the feature values and the service provider ID included in the authentication request and determines the corresponding service user ID (step **S14**).

[0167] The authentication server **10** transmits a response including the determined service user ID (a response to the authentication request) to the management server **20** (step **S15**).

[0168] The management server **20** searches its user information database by using the acquired service user ID and determines the corresponding personal information (step **S16**).

[0169] The management server **20** transmits the determined personal information to the authentication terminal **30** (step **S17**).

[0170] The authentication terminal **30** provides a service by using the acquired personal information (step **S18**).

[0171] As described above, in the authentication system according to the first example embodiment, while the biological information about the user is stored in the authentication server **10**, the biological information is not stored in the service provider. In addition, while the personal information about the user is stored in the management server **20** managed and operated by the service provider, the personal information is not stored in the authentication server **10**. In the authentication system according to the first example embodiment, the above items of information are stored at separate locations. Thus, the authentication system provides an authentication infrastructure that is robust against information leakage. That is, biological information (in particular, feature values) not associated with personal information is only a series of numerical values and is not valuable to criminals, etc. Thus, even if information is leaked from the authentication server **10**, the impact is limited. With the configuration as described above, the participants in the authentication system (the users receiving services and the service providers providing these services) can securely use the authentication system.

Second Example Embodiment

[0172] Next, a second example embodiment will be described in detail with reference to drawings.

[0173] In the first example embodiment, biological information used for the authentication of a user is stored in the authentication center, and personal information needed when the user uses a service is stored in a service provider. With this configuration, an authentication system highly resistant to information leakage (robust against information leakage) is established.

[0174] However, with the configuration according to the first example embodiment, there is still a possibility that the personal information is leaked from the service provider. In the second example embodiment, an authentication system highly resistant to information leakage from the service provider will be described.

[0175] As the authentication system according to the second example embodiment can have the same configuration as that according to the first example embodiment, the description corresponding to FIG. 2 will be omitted. In addition, as the authentication server **10**, the individual management server **20**, and the individual authentication terminal **30** according to the second example embodiment can have the same processing configurations as those according to first example embodiment, the description thereof will be omitted.

[0176] The following description will be made with a focus on the difference between the first and

second example embodiments.

[0177] The authentication system according to the second example embodiment differs from the authentication system according to the first example embodiment in the operations in the service registration phase and the service provision phase.

[0178] A schematic operation in the service registration phase according to the second example embodiment will be described with reference to FIG. 18. Upon receiving a service registration request from a management server 20, the authentication server 10 generates an encryption key. The generated encryption key is associated with the corresponding user ID, password, service provider ID, and service user

[0179] ID, and the associated data is stored in the authentication information database. The authentication server 10 transmits the generated encryption key and the service user ID to the management server 20. The management server 20 generates encrypted text Enc of the corresponding personal information acquired from the user by using the acquired encryption key and stores the encrypted personal information and the service user ID in association with each other. The management server 20 deletes the personal information in plain text.

[0180] A schematic operation in the service provision phase according to the second example embodiment will be described with reference to FIG. 19. Upon receiving an authentication request from a management server 20, the authentication server 10 searches its authentication information database and determines the corresponding service user ID and encryption key. The authentication server 10 transmits a response including the determined service user ID and encryption key (a response to the authentication request) to the management server 20. The management server 20 searches its user information database by using the acquired service user ID as a key and determines the corresponding personal information (encrypted personal information). The management server 20 decrypts the encrypted personal information by using the encryption key acquired from the authentication server 10. The management server 20 transmits the corresponding personal information in plain text to a corresponding authentication terminal 30. The authentication terminal 30 provides a service by using the personal information.

[0181] Next, detailed operations of the authentication server 10 and the individual management server 20 according to the second example embodiment will be described.

[0182] When the service registration unit 204 in the authentication server 10 receives a service registration request (a request including a user ID, a password, and a service provider ID) from a management server 20, the service registration unit 204 generates an encryption key. The encryption key is a key for encrypting the personal information entered by the user to the corresponding service provider and for decrypting the encrypted personal information. The service registration unit 204 generates an encryption key for an individual pair of a user and a service provider.

[0183] For example, the service registration unit 204 generates a common key, which is a key that enables both encryption and decryption. For example, the service registration unit 204 calculates a hash value, which is a value obtained by combining a user ID, a password, and a service provider ID. The service registration unit 204 generates a common key by using the calculated hash value as a “seed”.

[0184] The user ID and the password are each a value unique to each user, and the service provider ID is a value unique to each service provider. Thus, a hash value (a seed for generating a key) calculated from a combined value of these values is a value unique to a combination of a user and a service provider. An encryption key calculated from a “seed”, which is generated for each combination of a user and a service provider, is also a value unique to each combination of a user and a service provider.

[0185] The service registration unit 204 gives the generated encryption key, etc. to the database management unit 203.

[0186] The database management unit 203 updates the authentication information database by

using the acquired encryption key. For example, the authentication information database according to the second example embodiment includes an encryption key field as illustrated in FIG. 20.

[0187] As illustrated in FIG. 20, an encryption key K (U1S1) is generated for a combination of the user U1 and the service provider S1, and an encryption key K (U1S2) is generated for a combination of the user U1 and the service provider S2.

[0188] The service registration unit **204** transmits a response including the service user ID and the encryption key generated in response to the service registration request to the management server **20** that has transmitted the service registration request.

[0189] The service registration request unit **303** in the management server **20** receives the response to the service registration request. The service registration request unit **303** extracts the encryption key from the response and encrypts the personal information (for example, the name, etc.) of the user. The service registration request unit **303** gives the encrypted personal information and the service user ID to the database management unit **304**. Next, the service registration request unit **303** deletes the original personal information (the personal information in plain text) and the encryption key.

[0190] The database management unit **304** adds an entry including the encrypted personal information and the service user ID in the user information database. The user information database stores the encrypted personal information and the service user ID (a third ID) in association with each other.

[0191] When the authentication unit **205** of the authentication server **10** receives an authentication request (an authentication request including the feature values of a user and a service provider ID) from a management server **20**, the authentication unit **205** performs authentication processing by using these items of information. If the authentication succeeds, the service user ID and an encryption key that correspond to the combination of the user and the service provider are determined.

[0192] The authentication unit **205** transmits a response including the determined service user ID and encryption key (a response to the authentication request) to the management server **20**.

[0193] The authentication request unit **305** of the management server **20** searches the user information database by using the acquired service user ID as a key and acquires the corresponding personal information (encrypted personal information).

[0194] The authentication request unit **305** decrypts the encrypted personal information by using the encryption key (common key) acquired from the authentication server **10**. The authentication request unit **305** transmits the personal information in plain text to the authentication terminal **30**. If the authentication request unit **305** succeeds in decrypting the personal information, the authentication request unit **305** deletes (discards) the encryption key acquired from the authentication server **10**.

[0195] It is desirable that the authentication terminal **30** delete the personal information in plain text after providing a service to the user.

[0196] As described above, the authentication server **10** according to the second example embodiment generates an encryption key in response to a service registration request and transmits the generated encryption key and service user ID (third ID) to the corresponding management server **20**. The management server **20** encrypts the personal information of the user by using the acquired encryption key and stores the encrypted personal information and the service user ID in association with each other. In addition, the authentication server **10** receives an authentication request including biological information of a user and a service provider ID (second ID) from a service provider. The authentication server **10** determines a service user ID (third ID) and an encryption key by using the two items of biological information and a service provider ID. The authentication server **10** transmits the determined service user ID and encryption key to the corresponding service provider. When the corresponding management server **20** provides a service to the user, the management server **20** determines the encrypted personal information by using the

service user ID acquired by transmitting an authentication request to the authentication server **10**. In addition, the management server **20** decrypts the encrypted personal information by using the encryption key acquired from the authentication server **10**. The service is provided by using the decrypted personal information.

[0197] As described above, according to the second example embodiment, the personal information held by the individual service provider is encrypted. Even if the encrypted personal information is leaked, not a serious problem is caused. This is because security and privacy problems occur when the encrypted personal information is decrypted and the content thereof is acquired by a third party. The encryption keys to decrypt the encrypted personal information are stored in the authentication server **10**, and each time a service is provided to a user, an encryption key is transmitted from the authentication server **10** to the corresponding service provider. As the authentication system according to the second example embodiment is configured in this way, it is hard to assume a situation in which information leakage from a management server **20** and information leakage from the authentication server **10** occur simultaneously (during the same period of time). Thus, even if encrypted personal information is leaked, if the encryption key is protected by the authentication center, this information leakage will not be a serious problem. In addition, since the management server **20** deletes the encryption key after the personal information is decrypted, the service provider itself cannot access the personal information. Therefore, the employees, etc. of the service provider are prevented from improperly acquiring the personal information, for example.

Third Example Embodiment

[0198] Next, a third example embodiment will be described in detail with reference to drawings.

[0199] According to the second example embodiment, a service provider encrypts personal information, and the authentication server **10** holds an encryption key for decrypting the encrypted text. According to the second example embodiment, since the encrypted text and the encryption key are separately stored, an authentication system highly resistant to leakage of the personal information can be provided. However, for example, improvement in information processing technology could break the encryption of encrypted text leaked from a management server **20**. That is, it is not desirable for even encrypted personal information to be leaked from a service provider.

[0200] According to a third example embodiment, the authentication server **10** holds encrypted personal information, and the management servers **20** hold the corresponding encryption keys. In this way, an authentication system highly resistant to leakage of personal information can be provided.

[0201] As the authentication system according to the third example embodiment can have the same configuration as that according to the first example embodiment, the description corresponding to FIG. **2** will be omitted. In addition, as the authentication server **10**, the management servers **20**, and the authentication terminals **30** according to the third example embodiment can have the same processing configurations as those according to the first example embodiment, description thereof will be omitted.

[0202] Hereinafter, the third example embodiment will be described with a focus on the difference from the first and second example embodiments.

[0203] The authentication system according to the third example embodiment differs from those according to the first and second example embodiments in the operations in the service registration phase and the service provision phase.

[0204] A schematic operation in the service registration phase according to the third example embodiment will be described with reference to FIG. **21**. The authentication server **10** generates an encryption key for encrypting personal information of a user. The generated encryption key is transmitted to a management server **20**, along with a service user ID. The management server **20** generates encrypted text of the personal information by using the acquired encryption key and transmits a personal information registration request including the generated encrypted text (encrypted text Enc illustrated in FIG. **21**) and the service user ID to the authentication server **10**.

The authentication server **10** searches its authentication information database by using the service user ID as a key and determines a corresponding entry. The authentication server **10** stores the encrypted personal information in the determined entry. After transmitting the personal information registration request, the management server **20** deletes the personal information in plain text. In addition, the management server **20** stores the encryption key and the service user ID acquired from the authentication server **10** in association with each other in its user database.

[0205] A schematic operation in the service provision phase according to the third example embodiment will be described with reference to FIG. **22**. Upon receiving an authentication request, the authentication server **10** determines a corresponding service user ID and encrypted personal information by using its authentication information database. The authentication server **10** transmits a response (a response to the authentication request) including the determined service user ID and personal information to the management server **20**. The management server **20** searches its user information database by using the acquired service user ID as a key and determines a corresponding encryption key. The management server **20** decrypts the personal information acquired from the authentication server **10** by using the determined encryption key. The management server **20** transmits the personal information in plain text to an authentication terminal **30**. The authentication terminal **30** provides a service by using the personal information.

[0206] Next, detailed operations of the authentication server **10** and the individual management server **20** according to the third example embodiment will be described.

[0207] When the service registration unit **204** of the authentication server **10** receives a service registration request (a request including a user ID, a password, and a service provider ID) from a management server **20**, the service registration unit **204** generates an encryption key (a common key) according to the second example embodiment.

[0208] The service registration unit **204** transmits a response including a service user ID and an encryption key generated in response to the service registration request to the management server **20** that has transmitted the service registration request.

[0209] The service registration request unit **303** of the management server **20** receives a response to the service registration request. The service registration request unit **303** extracts the encryption key from the response and encrypts personal information (for example, the name, etc.) of the user. After encrypting the personal information, the service registration request unit **303** deletes the personal information in plain text.

[0210] The service registration request unit **303** generates a “personal information registration request” including the encrypted personal information and the service user ID and transmits the “personal information registration request” to the authentication server **10**.

[0211] In addition, the service registration request unit **303** gives the service user ID and the encryption key acquired from the authentication server **10** to the database management unit **304**. The database management unit **304** adds a new entry including the service user ID and the encryption key in the user information database (see FIG. **23**). As illustrated in FIG. **23**, the user information database stores the service user ID and the encryption key in association with each other.

[0212] The service registration unit **204** of the authentication server **10** receives the personal information registration request. The service registration unit **204** gives the personal information registration request to the database management unit **203**.

[0213] The database management unit **203** extracts the service user ID from the personal information registration request and searches the authentication information database by using the ID as a key. The database management unit **203** stores the encrypted personal information in the determined entry. As a result, for example, an authentication information database having a personal information field as illustrated in FIG. **24** is obtained. The authentication information database stores the encrypted personal information in association with the biological information of the user, the user ID, etc.

[0214] As illustrated in FIG. 24, encrypted text Enc (U11) of personal information entered by the user U1 to the service provider S1 and encrypted text Enc (U12) of personal information entered by the user U1 to the service provider S2 are registered in the database.

[0215] When the authentication unit **205** of the authentication server **10** receives an authentication request (an authentication request including feature values of the user and the service provider ID) from the management server **20**, the authentication unit **205** performs authentication processing by using these items of information. If the authentication unit **205** succeeds in the authentication, the service user ID and the encrypted personal information corresponding to the combination of the user and the service provider are determined.

[0216] The authentication unit **205** transmits a response (a response to the authentication request) including the determined service user ID and encrypted personal information to the management server **20**.

[0217] The authentication request unit **305** of the management server **20** searches the user information database by using the acquired service user ID as a key and determines a corresponding encryption key.

[0218] The authentication request unit **305** decrypts the encrypted personal information acquired from the authentication server **10** by using the encryption key. The authentication request unit **305** transmits the personal information in plain text to a corresponding authentication terminal **30**. If the authentication request unit **305** succeeds in decrypting the personal information, the authentication request unit **305** deletes (discards) the encrypted personal information acquired from the authentication server **10**.

[0219] As described above, in the authentication system according to the third example embodiment, the encrypted personal information is stored in the authentication server **10**, and the encryption key for decrypting the personal information is stored in the management server **20**. Normally, robust measures against unauthorized access and the like by a third party are taken for the authentication server **10** installed at an authentication center. In contrast, there are cases in which measures against the above unauthorized access and the like are insufficient for a management server **20** installed at a service provider. Thus, there is a difference in security strength between the authentication server **10** and the management server **20**, and personal information is stored in the authentication server **10** having a higher security strength (it is hard to assume that information will leak from the authentication server **10**). Thus, compared with the second example embodiment in which the management server **20** stores encrypted personal information, the risk of leakage of encrypted personal information is lower. In addition, while the management server **20** stores the encryption key and the service user ID, such information does not give any useful information to a third party. Thus, even if information is leaked from the management server **20** having a lower security strength, not a serious problem will be caused.

[0220] Next, a hardware configuration of an individual apparatus that constitutes the authentication system will be described. FIG. 25 is a diagram illustrating an example of a hardware configuration of the authentication server **10**.

[0221] The authentication server **10** can be configured by an information processing apparatus (a so-called computer) and has a configuration illustrated as an example in FIG. 25. For example, the authentication server **10** includes a processor **311**, a memory **312**, an input-output interface **313**, a communication interface **314**, etc. The components such as the processor **211** are connected to an internal bus, etc. so that these components can communicate with each other.

[0222] The hardware configuration of the authentication server **10** is not limited to the configuration illustrated in FIG. 25. The authentication server **10** may include hardware not illustrated or may be configured without the input-output interface **313** if desired. In addition, the number of components, such as the number of processors **311**, included in the authentication server **10** is not limited to the example illustrated in FIG. 25. For example, a plurality of processors **311** may be included in the authentication server **10**.

[0223] For example, the processor **311** is a programmable device such as a CPU (Central Processing Unit), an MPU (Micro Processing Unit), or a DSP (Digital Signal Processor). Alternatively, the processor **311** may be a device such as an FPGA (Field Programmable Gate Array) or an ASIC (Application Specific Integrated Circuit). The processor **311** executes various kinds of programs including an operating system (OS).

[0224] The memory **312** is a RAM (Random Access Memory), a ROM (Read-Only Memory), an HDD (Hard Disk Drive), an SSD (Solid State Drive), or the like. The memory **312** stores an OS program, an application program, and various kinds of data.

[0225] The input-output interface **313** is an interface for a display device and an input device not illustrated. For example, the display device is a liquid crystal display or the like. For example, the input device is a device, such as a keyboard or a mouse, which receives user operations.

[0226] The communication interface **314** is a circuit, a module, or the like for performing communication with other apparatuses. For example, the communication interface **314** includes a NIC (Network Interface Card) or the like.

[0227] The functions of the authentication server **10** are realized by various kinds of processing modules. The processing modules are realized, for example, by causing the processor **311** to execute a program stored in the memory **312**. In addition, this program can be recorded in a computer-readable storage medium. The storage medium may be a non-transient (non-transitory) storage medium, such as a semiconductor memory, a hard disk, a magnetic recording medium, or an optical recording medium. That is, the present invention can be embodied as a computer program product. In addition, the above program may be updated by downloading a program via a network or by using a storage medium in which a program is stored. In addition, the above processing modules may be realized by semiconductor chips.

[0228] The management servers **20**, the authentication terminals **30**, etc. can each be configured by an information processing apparatus, as in the case of the authentication server **10**. As each of these apparatuses can also have the same basic hardware configuration as that of the authentication server **10**, description thereof will be omitted. For example, the individual authentication terminal **30** includes a camera for capturing an image of an individual user.

[0229] The authentication server **10** includes a computer and can realize its functions by causing the computer to execute a program. In addition, the authentication server **10** executes an authentication server control method by using this program.

[Variations]

[0230] The configurations, operations, etc. of the authentication systems according to the above example embodiments are examples and do not limit the present system configuration, etc.

[0231] In the above example embodiments, a user (a system user) determines a user ID and a password, and this user registered in the system is determined by the user ID and the password. However, the authentication system may determine an ID (identifier) that uniquely determines the system user. For example, in the user registration phase, the authentication server **10** acquires biological information (a face image or feature values) of a user. The authentication server **10** may generate an ID based on the biological information. For example, the authentication server **10** may calculate a hash value from feature values of a face image and may use the calculated hash value in place of the user ID and the password. Feature values of a face image differ depending on the user, and a hash value generated from these feature values also differs depending on the user. Thus, the hash value can be used as the ID of the system user.

[0232] In the above example embodiments, the user registration phase and the service registration phase are performed at different timings. However, these phases may be performed at substantially the same timing. For example, an authentication terminal **30** installed at a service provider from which a user wishes to receive a service may be used for the above registration phases. Specifically, a user may use an authentication terminal **30** to perform user registration (entering biological information, a user ID, and a password) and to continuously perform service registration (entering

personal information, the user ID, and the password). In this case, the authentication terminal **30** includes the user registration function (the user registration unit **202**) of the authentication server **10** and the personal information acquisition function (the personal information acquisition unit **302**) of the management server **20**.

[0233] A plurality of authentication terminals **30** of a service provider may not need to be installed on the same premises, in the same building, etc. If a plurality of authentication terminals **30** are managed and operated by the same service provider, these authentication terminals **30** may be installed at locations spatially away from each other.

[0234] In the above example embodiments, a single service provider ID is assigned to a single service provider. However, a single service provider ID may be assigned to a plurality of service providers. That is, a plurality of service providers may be managed as a group, and a service provider ID may be issued per group. For example, in a case in which the service providers S1 and S2 coordinate with each other and provide the same service, a shared service provider ID may be issued to these service providers S1 and S2.

[0235] In the above example embodiments, biological information relating to “feature values generated from a face image” is transmitted from a management server **20** to the authentication server **10**. However, biological information relating to “a face image” may be transmitted from a management server **20** to the authentication server **10**. In this case, the authentication server **10** generates feature values from the acquired face image and performs authentication processing (matching processing).

[0236] In the above example embodiments, an authentication terminal **30** acquires a face image, and the corresponding management server **20** generates feature values from the face image. However, the authentication terminal **30** may generate feature values from a face image and may transmit the generated feature values to the corresponding management server **20**. That is, the management server **20** does not need to generate the feature values.

[0237] In the above example embodiments, in the service registration phase, when a user registers his or her personal information, the user enters a user ID and a password to a service provider (see FIG. **13**). However, in place of the user ID and the password, biological information (a face image) of the user may be entered to the service provider. In this case, the corresponding management server **20** transmits a service registration request including feature values generated from the face image and its service provider ID to the authentication server **10**. The authentication server **10** performs matching processing by using the feature values included in the request and the feature values registered in its authentication information database and determines the corresponding user. If the authentication server **10** succeeds in the user determination (authentication), the authentication server **10** issues a service user ID. In this way, even if the user forgets his or her user ID or password, the user can easily perform the service registration. Alternatively, the service provider may acquire biological information (a face image) of a user, in addition to the user ID and the password. In this case, if the user ID, password, and biological information match, the authentication server **10** may issue a service user ID (two-factor authentication using the biological information and password may be performed).

[0238] A service provider may cache (temporarily store) information acquired from the authentication server **10** or information acquired from an authentication terminal **30**. For example, a management server **20** caches biological information acquired from an authentication terminal **30** and an authentication result (a service user ID) based on the biological information for a predetermined period. Upon acquiring biological information from an authentication terminal **30**, first the corresponding management server **20** checks the data cached therein. If there is cached data that matches the acquired biological information, the management server **20** does not transmit an authentication request to the authentication server **10**. The management server **20** determines the corresponding personal information by using the service user ID included in the cached data. Alternatively, the management server **20** may cache a combination of biological information and

personal information. Alternatively, depending on the kind of the service, a different condition for deleting cached data may be used. For example, if an accommodation service is provided by a hotel company, the management server **20** may delete corresponding cached data when the duration of the stay of the guest ends.

[0239] In the above example embodiments, while the authentication server **10** generates an encryption key, the individual management server **20** may generate an encryption key. In this case, in the case of the second example embodiment, a management server **20** generates an encryption key and encrypts personal information. The management server **20** stores the encrypted personal information and the corresponding service user ID in association with each other. Next, the management server **20** transmits the encryption key to the authentication server **10**. The authentication server **10** stores the acquired encryption key in association with the corresponding user ID, etc. In the case of the third example embodiment, a management server **20** transmits the encrypted personal information to the authentication server **10**. The management server **20** stores the generated encryption key and the corresponding service user ID in association with each other. The authentication server **10** stores the acquired personal information (the encrypted text of the personal information), the corresponding user ID, etc. in association with each other. The management server **20** deletes the original personal information and the encryption key at the timing described in the second or third example embodiment. Since an individual management server **20** generates an encryption key, the number of communications between the authentication server **10** and the management server **20** can be reduced in the case of the third example embodiment.

[0240] In the above example embodiments, a common key is used as an encryption key for encrypting personal information. However, “private key” and “public key” may be used as the encryption key for encrypting personal information. In this case, the authentication server **10** stores a private key in association with a user ID, etc. In addition, the authentication server **10** gives a public key to an individual management server **20**. The management server **20** uses the public key to encrypt personal information. Upon receiving an authentication request, the authentication server **10** transmits the corresponding service user ID and private key to the management server **20**. The management server **20** decrypts the personal information by using the acquired private key.

[0241] Data transmission and reception modes among the individual apparatuses (the authentication server **10**, the individual management server **20**, and the individual authentication terminal **30**) are not limited to any particular modes. For example, data transmitted or received among these apparatuses may be encrypted. Among these apparatuses, biological information is transmitted and received. Thus, to appropriately protect the biological information, it is desirable that encrypted data be transmitted and received.

[0242] In the flowcharts and sequence diagrams used in the above description, a plurality of steps (processes) are sequentially described. However, the order of the execution of the steps performed in the individual example embodiment is not limited to the described order. In the individual example embodiment, the order of the illustrated steps may be changed to the extent that a problem is not caused on the content of the individual example embodiment. For example, individual processes may be executed in parallel.

[0243] The above example embodiments have been described in detail to facilitate the understanding of the present application disclosed and not to mean that all the configurations described above are needed. In addition, if a plurality of example embodiments have been described, each of the example embodiments may be used individually or a plurality of example embodiments may be used in combination. For example, part of a configuration according to one example embodiment may be replaced by a configuration according to another example embodiment. For example, a configuration according to one example embodiment may be added to a configuration according to another example embodiment. In addition, addition, deletion, or replacement is possible between part of a configuration according to one example embodiment and

another configuration.

[0244] The industrial applicability of the present invention has been made apparent by the above description. That is, the present invention is suitably applicable, for example, to authentication systems for authentication of customers of retail stores, hotel operators, etc.

[0245] A part or the entirety of the example embodiments described above may be described as in the following supplementary notes, but is not limited to the followings.

[Supplementary Note 1]

[0246] An authentication server including: [0247] a user registration unit that acquires a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; [0248] a service registration unit that processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; and [0249] a storage unit, [0250] wherein the service registration unit generates a third ID that is uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider, and [0251] wherein the storage unit stores the first biological information, the first ID, the second ID, and the third ID in association with each other.

[Supplementary Note 2]

[0252] The authentication server according to supplementary note 1, further including an authentication unit that receives an authentication request including second biological information of the user and the second ID from the service provider, determines the third ID by using the first and second biological information and the second ID, and transmits the determined third ID to the service provider.

[Supplementary Note 3]

[0253] The authentication server according to supplementary note 1 or 2, wherein the user registration unit acquires a password of the user, and the storage unit stores the first biological information, the first ID, the password, the second ID, and the third ID in association with each other.

[Supplementary Note 4]

[0254] The authentication server according to supplementary note 3, wherein the service registration unit calculates a hash value by using the first ID, the password, and the second ID and uses the calculated hash value as the third ID.

[Supplementary Note 5]

[0255] The authentication server according to any one of supplementary notes 1 to 4, wherein the first biological information includes a feature value generated from a face image of the user.

[Supplementary Note 6]

[0256] An authentication system including: [0257] an authentication server; and [0258] a management server, [0259] wherein the authentication server includes [0260] a user registration unit that acquires a first ID that uniquely determines a user in the system and first biological information that is used for authentication of the user, [0261] a service registration unit that processes a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider, and [0262] a first storage unit, [0263] wherein the service registration unit generates a third ID that is uniquely determined by a combination of the user and the service provider and transmits the third ID to the service provider, [0264] wherein the first storage unit stores the first biological information, the first ID, the second ID, and the third ID in association with each other, and [0265] wherein the management server includes [0266] a personal information acquisition unit that acquires the first ID and personal information of the user from the user, [0267] a service registration request unit that acquires the third ID by transmitting the service registration request to the authentication server, and [0268] a second storage unit that stores personal information of the user and the third ID in association with each other.

[Supplementary Note 7]

[0269] The authentication system according to supplementary note 6, [0270] wherein the authentication server further includes an authentication unit that receives an authentication request including second biological information of the user and the second ID from the service provider, determines the third ID by using the first and second biological information and the second ID, and transmits the determined third ID to the service provider, and [0271] wherein the management server further includes an authentication request unit that determines personal information of the user by using the third ID acquired by transmitting the authentication request to the authentication server when a service is provided to the user.

[Supplementary Note 8]

[0272] The authentication system according to supplementary note 7, further including an authentication terminal that acquires personal information of the user from the management server by transmitting biological information acquired from the user to the management server and that provides a service to the user by using the acquired personal information.

[Supplementary Note 9]

[0273] The authentication system according to supplementary note 8, [0274] wherein the authentication server stores a first feature value generated from a face image of the user as the first biological information, [0275] wherein the authentication terminal transmits the face image of the user to the management server, and [0276] wherein the management server uses a second feature value generated from the face image as the second biological information.

[Supplementary Note 10]

[0277] The authentication system according to any one of supplementary notes 6 to 9, wherein the personal information does not include biological information of the user.

[Supplementary Note 11]

[0278] A control method of an authentication server, the control method including: [0279] acquiring a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; [0280] receiving a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; [0281] generating a third ID that is uniquely determined by a combination of the user and the service provider; [0282] transmitting the third ID to the service provider; and storing the first biological information, the first ID, the second ID, and the third ID in association with each other.

[Supplementary Note 12]

[0283] A computer-readable storage medium storing a program causing a computer mounted on an authentication server to perform processing for: acquiring a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; [0284] receiving a service registration request that is transmitted from a service provider of a service that the user wishes to use and that includes the first ID and a second ID that identifies the service provider; [0285] generating a third ID that is uniquely determined by a combination of the user and the service provider; [0286] transmitting the third ID to the service provider; and [0287] storing the first biological information, the first ID, the second ID, and the third ID in association with each other.

[0288] The entire disclosure of the above patent literature is incorporated herein by reference thereto. While the example embodiments of the present invention have thus been described, the present invention is not limited to these example embodiments. It is to be understood to those skilled in the art that these example embodiments are only examples and that various variations are possible without departing from the scope and spirit of the present invention. That is, the present invention of course includes various variations and modifications that could be made by those skilled in the art in accordance with the overall disclosure including the claims and the technical concept.

REFERENCE SIGNS LIST

[0289] **10, 100** authentication server [0290] **20** management server [0291] **30** authentication terminal [0292] **40** terminal [0293] **101, 202** user registration unit [0294] **102, 204** service registration unit [0295] **103, 206, 306, 405** storage unit [0296] **201, 301, 401** communication control unit [0297] **203, 304** database (DB) management unit [0298] **205** authentication unit [0299] **302** personal information acquisition unit [0300] **303** service registration request unit [0301] **305** authentication request unit [0302] **311** processor [0303] **312** memory [0304] **313** input-output interface [0305] **314** communication interface [0306] **402** biological information acquisition unit [0307] **403** service provision unit [0308] **404** message output unit

Claims

1. An authentication server comprising: at least one memory storing a set of instructions; and at least one processor configured to execute the set of instructions to: acquire a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; and process a service registration request that includes the first ID and a second ID that identifies a service that the user wishes to use, wherein the at least one processor is further configured to execute the set of instructions to: generate a third ID that is uniquely determined by a combination of the user and the service; and store the first biological information, the first ID, the second ID, and the third ID in association with each other, and wherein storing the first biological information, the first ID, the second ID, and the third ID in association with each other comprises storing the biological information, the first ID, the second ID, and the third ID in a same table.
2. The authentication server according to claim 1, wherein the at least one processor is further configured to execute the set of instructions to: receive an authentication request including second biological information of the user and the second ID from a service provider; determine the third ID by using the first and second biological information and the second ID; and transmit the determined third ID to the service provider.
3. The authentication server according to claim 1, wherein the at least one processor is further configured to execute the set of instructions to: acquire a password of the user; and store the first biological information, the first ID, the password, the second ID, and the third ID in association with each other.
4. The authentication server according to claim 3, wherein the at least one processor is further configured to execute the set of instructions to: calculate a hash value by using the first ID, the password, and the second ID; and use the calculated hash value as the third ID.
5. The authentication server according to claim 1, wherein the first biological information includes a feature value generated from a face image of the user.
6. An authentication system comprising: an authentication server; and a management server, wherein the authentication server includes at least one memory storing a set of instructions; and at least one processor configured to execute the set of instructions to: acquire a first ID that uniquely determines a user in the system and first biological information that is used for authentication of the user; process a service registration request that includes the first ID and a second ID that identifies a service that the user wishes to use, wherein the at least one processor is further configured to execute the set of instructions to: generate a third ID that is uniquely determined by a combination of the user and the service; and store the first biological information, the first ID, the second ID, and the third ID in association with each other, wherein storing the first biological information, the first ID, the second ID, and the third ID in association with each other comprises storing the biological information, the first ID, the second ID, and the third ID in a same table, and wherein the management server includes at least one memory storing a set of instructions; and at least one processor configured to execute the set of instructions to: acquire the first ID and personal information of the user from the user; acquire the third ID by transmitting the service registration

request to the authentication server; and store personal information of the user and the third ID in association with each other.

7. The authentication system according to claim 6, wherein the at least one processor of the authentication server is further configured to execute the set of instructions to: receive an authentication request including second biological information of the user and the second ID from a service provider; determine the third ID by using the first and second biological information and the second ID; and transmit the determined third ID to the service provider, and wherein the at least one processor of the management server is further configured to execute the set of instructions to: determine personal information of the user by using the third ID acquired by transmitting the authentication request to the authentication server when a service is provided to the user.

8. The authentication system according to claim 7, further comprising an authentication terminal that acquires personal information of the user from the management server by transmitting biological information acquired from the user to the management server and that provides the service to the user by using the acquired personal information.

9. The authentication system according to claim 8, wherein the authentication server stores a first feature value generated from a face image of the user as the first biological information, wherein the authentication terminal transmits the face image of the user to the management server, and wherein the management server uses a second feature value generated from the face image as the second biological information.

10. The authentication system according to claim 6, wherein the personal information does not include biological information of the user.

11. A control method of an authentication server, the control method comprising: acquiring a first ID that uniquely determines a user in a system and first biological information that is used for authentication of the user; receiving a service registration request that includes the first ID and a second ID that identifies a service that the user wishes to use; generating a third ID that is uniquely determined by a combination of the user and the service; and storing the first biological information, the first ID, the second ID, and the third ID in association with each other, wherein storing the first biological information, the first ID, the second ID, and the third ID in association with each other comprises storing the biological information, the first ID, the second ID, and the third ID in a same table.

12. The authentication server according to claim 1, wherein generating the third ID comprises combining the first ID and the second ID as the combination.

13. The authentication server according to claim 1, wherein the at least one processor is further configured to execute the set of instructions to, based on the third ID, process implementation of a second service, other than the service, requested by the user.
